

Final EDA Report - Used Car Pricing Regression

05/13/2026

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1. Introduction

1.1 Inspiration

Our inspiration for exploring this dataset stemmed from a profound interest in cars and the used car market. One group member, Ben, is quite the enthusiast for the automotive industry, and proposed working around the topic. What's interesting about the used car market is how rapidly prices have inflated in recent years. The average sale price of a used car in the United States is up to over \$28,000 this year, so we thought it would be interesting to figure out primarily which factors drive used car prices to better predict market value for cars of our own.

1.2 Predictive Task

With this model, we wanted to figure out what factors best predict the selling price of a used car. Although our data source only provides us with the listing price, we can consider this as somewhat equivalent to the market value sale price, though it is likely somewhat inflated with many outliers.

1.3 Summary of Findings

- The Random Forest model produced the strongest predictive performance, explaining roughly 80% of the variance in used car prices.
- Vehicle age, manufacturing year, and odometer mileage were the most influential predictors of price.
- Significant outliers in Craigslist listing prices initially caused extremely poor initial model performance, making data cleaning an essential step in the modeling process.
- Residual analysis suggested heteroscedasticity, with prediction errors increasing for very low and very high-priced vehicles.

2. Sourcing

2.1 Data Source

Our data is sourced from a Kaggle dataset on used car prices. The data was collected from web scraping Craigslist's used car marketplace, and contains over four hundred thousand entries.

Further information on the dataset can be found here:

<https://www.kaggle.com/datasets/austinreese/craigslist-carstrucks-data>

2.2 Primary Input - *vehicles.csv*

Our main source of actual data was the comma-separated value file from this Kaggle source. As mentioned above, this included information about over four hundred thousand used car listings across the United States posted on Craigslist. The initial dataset contains 26 rows, though in order to make the data more meaningful for regression analysis, we had to drop a few useful rows and add a few that we engineered as well. Since this dataset was huge, we took about 10% of it for sampling. This is because trying to fit models on the full 426,000-entry dataset, even in a high-RAM runtime connected to stronger GPUs, led to the program timing out.

2.3 Variables

1. *id*
 - a. Integer. Assigned ID of the vehicle in the dataset. This column was dropped because it would not be meaningfully predictive of price.
2. *Url*
 - a. Object. URL link to the original listing. This column was dropped because it would not be meaningfully predictive of price.

3. *Region*

- a. Object. Craigslist region in which the listing originated. Although likely predictive, this column contained far too many unique values to OHE, so it had to be dropped.

4. *Region_url*

- a. Object. URL link to the region's homepage. This column was dropped because it would not be meaningfully predictive of price.

5. *Price*

- a. Integer. Listed price for the vehicle up for sale. This is the “target” variable that we hope to predict. This data is heavily skewed and requires cleaning.

6. *Year*

- a. Float. Production year of the vehicle.

7. *Manufacturer*

- a. Object. Manufacturer of the vehicle.

8. *Model*

- a. Object. Specific model of the vehicle. Even though this column is likely predictive, it has countless unique values, and was leading to difficulties in the fitting of models because even with only 42000 datapoints, there were over 7000 unique models. Therefore, this column was dropped.

9. *Condition*

- a. Object. Categorical description of the condition of the vehicle (new, like new, excellent, good, fair, salvage)

10. *Cylinders*

- a. Object. String describing the number of cylinders the vehicle's engine has.

11. *Fuel*

- a. Object. Categorical description of the type of fuel used by the vehicle (gas, diesel, hybrid, electric, or other).

12. Odometer

- a. Float. Listed the odometer reading for the car.

13. Title_status

- a. Object. Categorical description of the title of the car (clean, rebuilt, salvage, lien, missing, parts only).

14. Transmission

- a. Object. Categorical description of the transmission of the vehicle (automatic, manual, other).

15. VIN

- a. Object. String vehicle identification number. This column was dropped because it would not be meaningfully predictive of price.

16. Drive

- a. Object. Drive type for the vehicle (4WD, FWD, RWD).

17. Size

- a. Object. Categorical descriptor for vehicle size (full-size, mid-size, compact, sub-compact)

18. Type

- a. Object. Categorical descriptor for type of vehicle (ex. Coupe, truck, sedan, etc.).

19. Paint_color

- a. Object. Description of the paint color of the vehicle.

20. Image_url

- a. Object. URL link to the listing images. This column was dropped because it would not be meaningfully predictive of price.

21. *Description*

- a. Object. Full listing description. This column was dropped because it would not be meaningfully predictive of price.

22. *County*

- a. Float, for some odd reason. County in which the vehicle was listed. This column was dropped because it was completely empty.

23. *State*

- a. Object. State in which the vehicle was listed.

24. *Lat*

- a. Float. Latitude of the listing.

25. *Long*

- a. Float. Longitude of the listing.

26. *Posting_date*

- a. Object. Date on which the listing was posted. This was converted to a datetime column.

2.4 Feature Engineering

1. *vehicle_age*

- a. Float. Age of the car, calculated by subtracting the model year from the year in which it was sold.

2. *miles_per_year*

- a. Float. Number of miles travelled per year by the vehicle, which may tell us more about the wear and tear. Calculated by dividing the odometer reading by vehicle age.

3. *is_luxury*

- a. Integer. Binary categorical variable. Whether or not the car is considered a luxury vehicle (ex. German import, luxury Japanese, etc.)

3. Methodology

To explore this data set, we used a simple imputer to fill missing values, with categorical values using the most frequent strategy, and numerical values using the median due to skew in the dataset. We then made a transformer that applied the one-hot encoder to the categorical features, and a StandardScaler to the numerical features to get the data properly ready for models. As the data set was huge, we decided to use 10% of the data among 4 models: linear regression, decision tree, k-nearest neighbor, and random forest. We then used a grid search to find the best parameters for the best possible R^2 model.

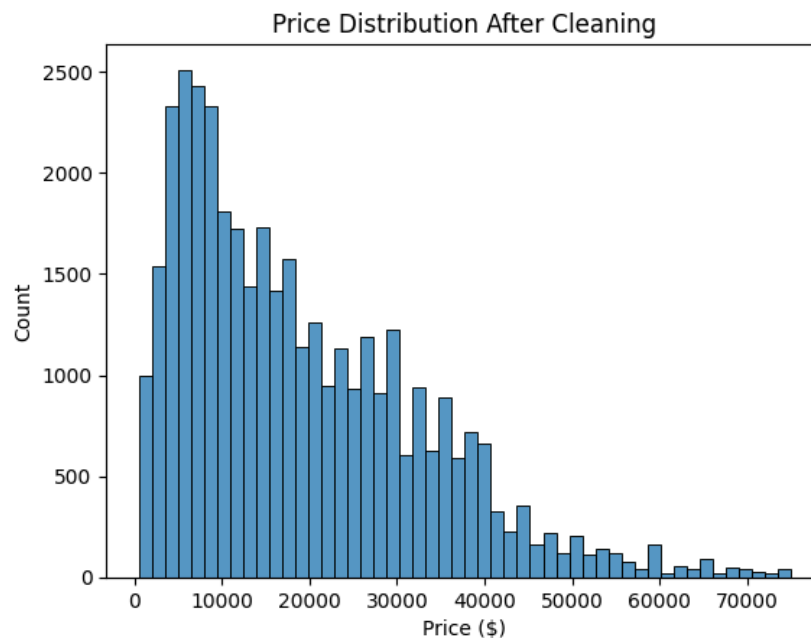
4. Exhibits and Interpretation

Fig. 4.1: *Initial Fitting of the Data*

Linear Regression
R^2 : -158.5724
MAE: 125,095
RMSE: 199,195
KNN
R^2 : -69769.5169
MAE: 85,003
RMSE: 4,165,195
Decision Tree
R^2 : -1.3842
MAE: 6,935
RMSE: 24,349
Random Forest
R^2 : -123045.0656
MAE: 65,370
RMSE: 5,531,376

A super negative R-squared indicates large price outliers. These massively negative R-squared values mean that high price outliers are destroying the models. This is likely because people will post absurdly high prices on their listings, potentially in the hundreds of millions of dollars, for no reason at all. For this reason, we have to examine the price more closely and make some adjustments to improve the fit of our model.

Fig. 4.2: *New Price Distribution Post-Cleaning*



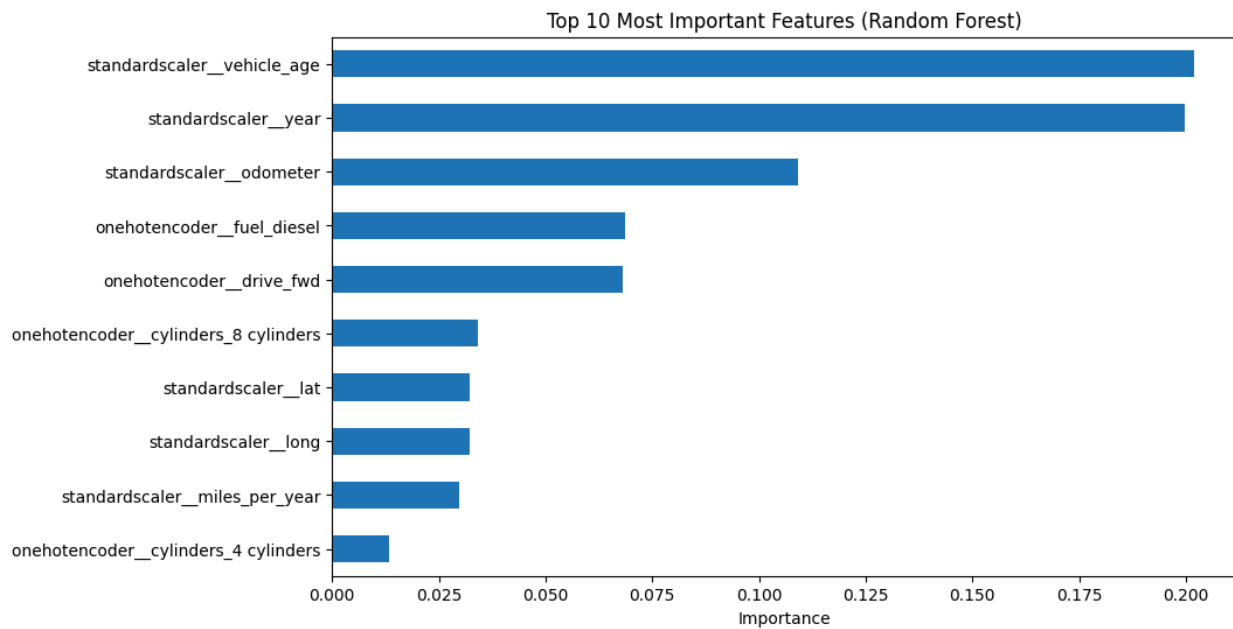
In order to eliminate a large number of outliers, we trimmed any data on cars with prices below \$750 or above \$75,000. In doing so, we hoped to remove most of the outliers skewing the data while also preserving the overall quality of the model. Clearly, after cleaning the data, the distribution of price is much more reasonable for fitting a model.

Fig 4.3: *Post-Cleaning Model Performance*

Linear Regression	
R ² :	0.4541
MAE:	7,259
RMSE:	10,208
KNN	
R ² :	0.6688
MAE:	5,341
RMSE:	7,951
Decision Tree	
R ² :	0.6335
MAE:	4,664
RMSE:	8,364
Random Forest	
R ² :	0.8046
MAE:	3,649
RMSE:	6,108

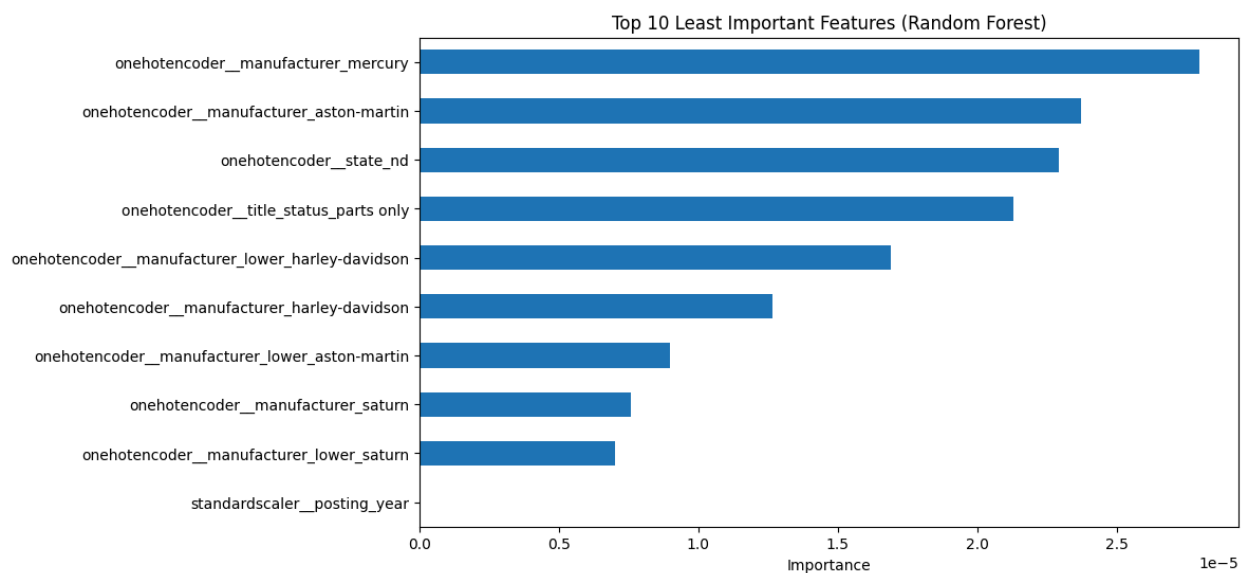
The model performance increased drastically post-cleaning, as it was primarily demonstrated in the predictive models' respective R² values. Each explains a significant portion of the data, and Random Forest explaining 80% exhibits the model's ability to predict price to a significant degree. Therefore, we decided to move forward with the random forest model as our central modeling method due to its high predictive power.

Fig 4.4: *Most Important/Predictive Features - Random Forest*



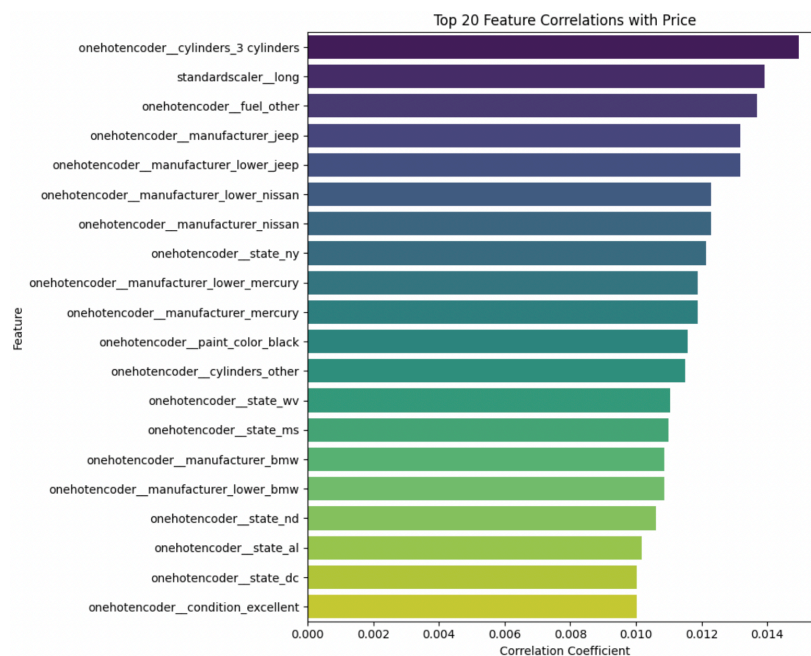
To understand this explanation of price, we looked at the specific primary features, with age, year, and miles (which makes sense) being the strongest. These values also demonstrate the desires of the buyer in pricing, and potentially the perspective on older cars, regardless of condition or brand.

Fig 4.5: *Least Important/Predictive Features - Random Forest*



Similar to the above, these features are the least valuable or important to predicting price in the Random Forest model, wherein consumer behavior is again demonstrated. The lack of presence, and frequency of presence, in the most and least important features, respectively, of the manufacturer type illustrates that other conditions/features are far more valued. This, to us, was surprising, as we believed the brand would make a significant impact on price, and be an obvious predictor factor that we recognized. This trend is likely because whether or not a car is from a sole manufacturer alone will likely be less impactful on price than all of the other variables at play.

Fig 4.6: *Correlation Matrix - Feature Against Price*



Respective values:

Top 10 features most positively correlated with price:

price	1.000000
onehotencoder__cylinders_3 cylinders	0.014952
standardscaler__long	0.013907
onehotencoder__fuel_other	0.013689
onehotencoder__manufacturer_jEEP	0.013191
onehotencoder__manufacturer_lower_jEEP	0.013191
onehotencoder__manufacturer_lower_nissan	0.012288
onehotencoder__manufacturer_nissan	0.012288
onehotencoder__state_ny	0.012126
onehotencoder__manufacturer_lower_mercury	0.011882
onehotencoder__manufacturer_mercury	0.011882

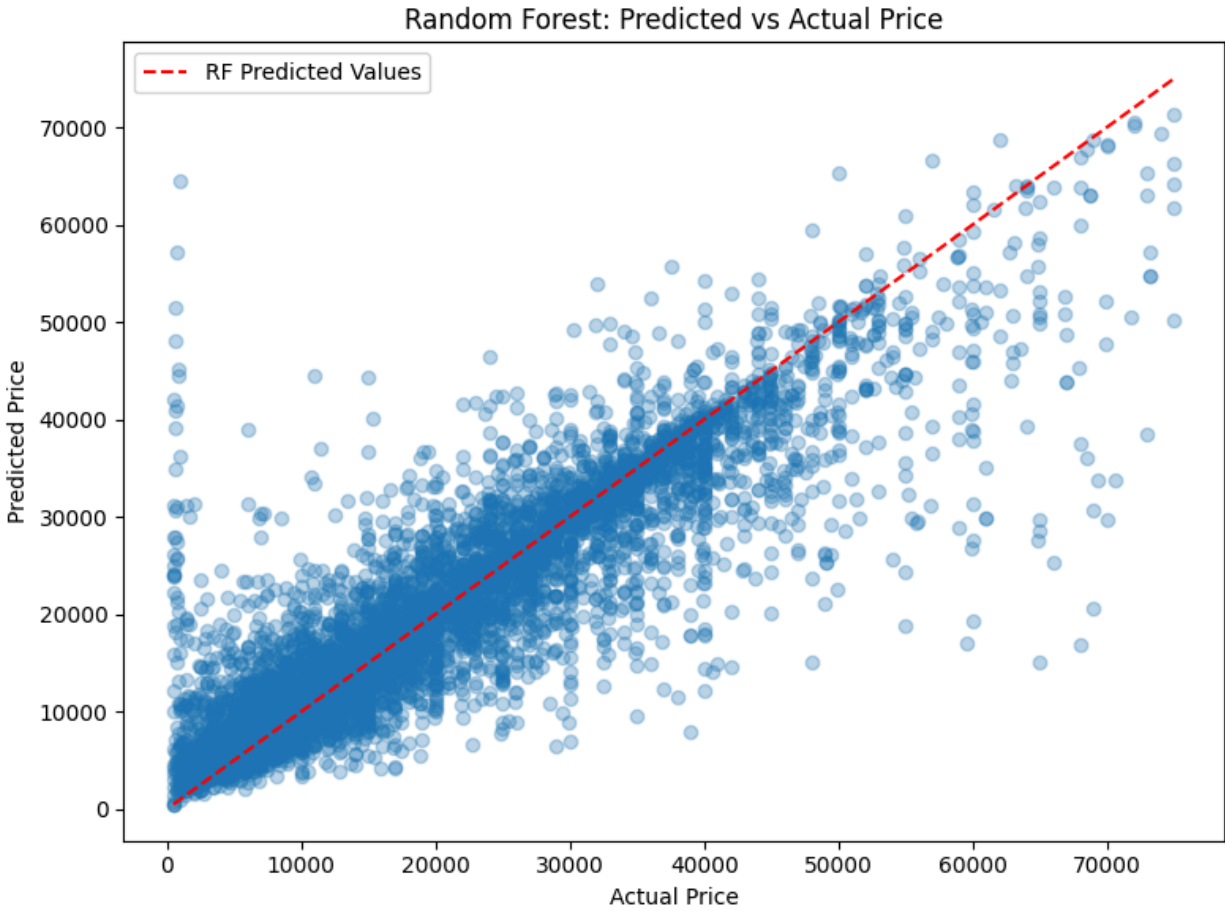
Name: price, dtype: float64

Top 10 features most negatively correlated with price:

onehotencoder__fuel_diesel	-0.011053
onehotencoder__type_bus	-0.011228
onehotencoder__manufacturer_lexus	-0.011394
onehotencoder__manufacturer_lower_lexus	-0.011394
onehotencoder__manufacturer_saturn	-0.011577
onehotencoder__manufacturer_lower_saturn	-0.011577
onehotencoder__state_ca	-0.011862
standardscaler__miles_per_year	-0.015250
onehotencoder__state_co	-0.021249

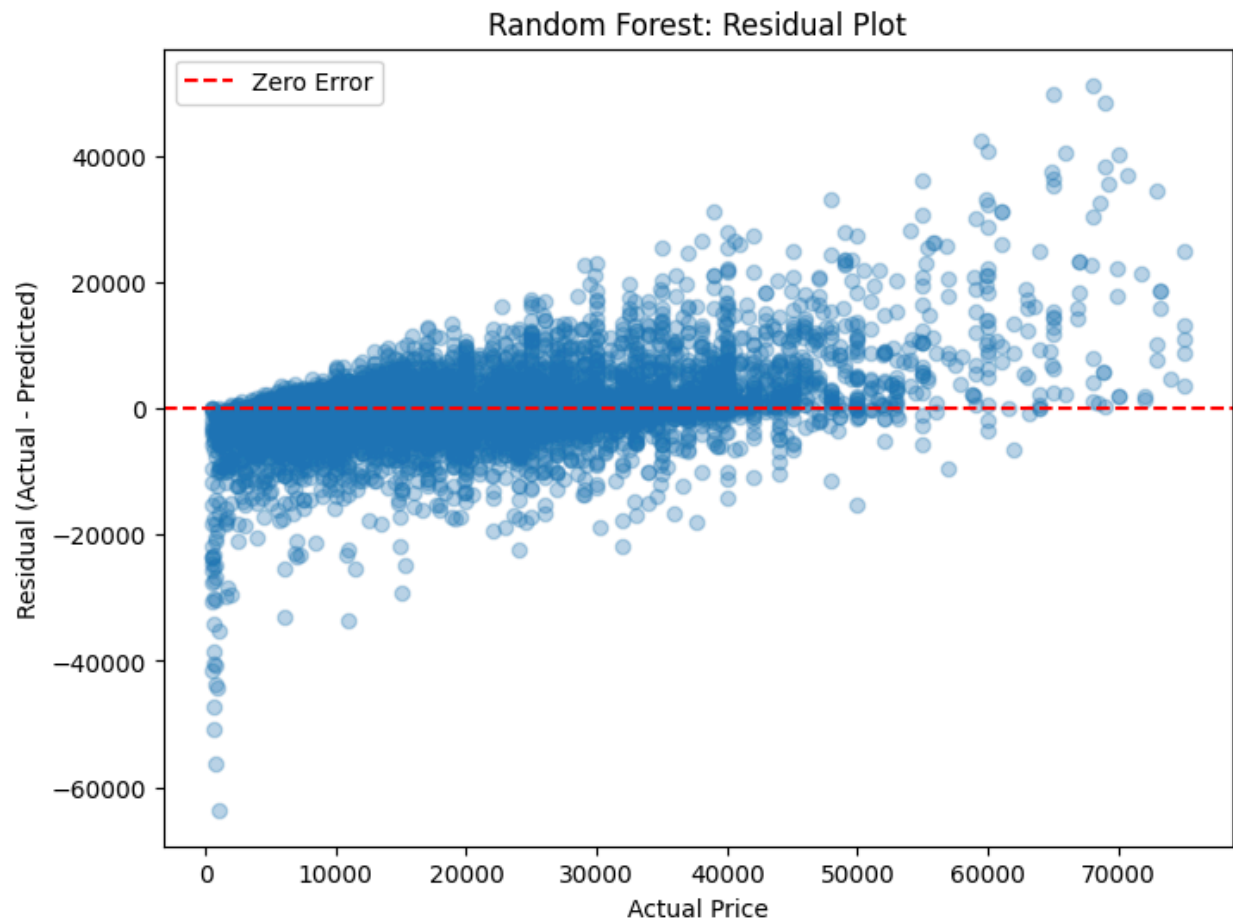
No one feature demonstrated a significant positive or negative linear correlation to the target variable, price, suggesting the need for these other predictive models to explain the complexity of pricing dynamics. To suggest this, we can look at this figure next to Figures 4.4 and 4.5, to show that features, even those with low linear correlation, are indeed important for the model's overall predictive power due to their roles in non-linear interactions.

Fig 4.7: *Model Performance - Predicted Price vs. Actual Price*



This 80% of explained data is visibly evident within this graph, where the model performance vs actual price is highly concentrated on the exact correlation line, suggesting that although it is not perfect, as a predictive model with a smaller portion of the data, we can learn and confidently predict the price of the cars.

Fig 4.8: *Model Performance - Residual Plot*



Further validating this above point is that most of the residuals sit on or around 0 residuals between the real and predicted, with an important note to make: Extremely low residuals were recognized at very low prices, and very high residuals were recognized at higher prices. This is most likely due to the concept of heteroscedasticity – that there are fewer examples/data points at these price points – and that due to this lack of sample size, the predictive quality drastically decreases. As for the extreme negative residuals, they can be explained by the fact that a lot of these postings were for down payments, and are thus outliers in the predictive model sense of this data.

5. Discussion

In terms of discussion criteria to beware of when looking at this predictive model, there are a few things to note. We initially had to drop two columns (region and model), as we had too many unique variables to one-hot-encode, and we believed these two were not necessarily as important to the predictive model, and were limited by the hardware of RAM and processing capabilities. This is also evident when you run our Colab, as it takes some patience to wait for the code to run on some fitting cells.

Another important aspect to note is the necessity of subsetting down to 10% of our original data, which most likely lacked any direct impact on our findings or the quality of our predictive model, due to the immense size of the data set. On top of this, the fact that we pulled this data from Craigslist resulted in some unregulated market activities (lack of price floor/ceiling), like rent/down payment prices as listings, and non-genuine, excessive asking prices.

6. Conclusion

In conclusion, the Random Forest model was the best choice due to its display in the R^2 value relative to the other predictive models. Through more analysis on Random Forest, we found that the most impactful variables were the vehicle's age, year manufactured, and odometer (how many miles were on it), and the least important were almost all having to do with manufacturers (Mercury and Aston Martin). Between price points \$10,000 and \$40,000, our model performed extremely well, as it was put onto an unregulated and thus unpredictable market; it was difficult to account for certain qualities in the data, as shown in our residuals graph. For instance, people inflate prices, and the listing price was too high, from \$40,000 onwards, compared to the predicted price. Sometimes, companies post cars as rentals or list only the down payment price, so a lot of cars listed

at \$2000 and below are not actually for sale for that price, meaning that residuals would be super negative. We dealt with this by excluding super low prices below \$750 to avoid much of this, yet some still got in. In addition, we had to avoid high prices that were skewing the dataset, set as unrealistic and non-serious (relative to real market pricing).

Regarding next steps in training this model, it would be interesting to compare it against a dataset of recorded car sale prices to test its predictive value. While the training on the large Craigslist data set is likely strong, the target is again based on listing prices rather than verified sale prices. Otherwise, it may be interesting to examine other important factors for a car's value that were not included in the dataset, especially numeric variables such as the number of owners or the number of reported accidents